Implementation of FastSpeech 2-Based Text-to-Speech Synthesis for English and Telugu

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1. Implementation approach

Two independent FastSpeech 2 acoustic models were built: one for English, trained on the Svarah corpus of Indian-accented English, and one for Telugu, trained on IndicTTS Telugu. Both follow Ren et al., *FastSpeech 2: Fast and High-Quality End-to-End Text to Speech* (ICLR 2021). They share language-independent library code (`fs2/`). Each language has its own text frontend, vocabulary, `config.yaml`, data, normalisation statistics and checkpoints, and neither pipeline reads anything from the other. Waveforms are generated with the pretrained universal HiFi-GAN vocoder (22.05 kHz, 80 mel bins).

1.1 Reference implementation and reused components

The base is **ming024/FastSpeech2** (MIT licence, github.com/ming024/FastSpeech2). The following components are reused with the same structure and hyper-parameters:

- **Encoder:** phoneme embedding, sinusoidal positional encoding, and 4 Feed-Forward Transformer (FFT) blocks. Each block has 2-head self-attention (hidden size 256) and a two-layer 1-D convolution feed-forward network (1024 filters, kernels 9 and 1), with residual connections, layer normalisation and dropout 0.2.
- **Variance adaptor:** duration, pitch and energy predictors, each made of 2×[Conv1D (kernel 3, 256 filters) + ReLU + LayerNorm + Dropout 0.5] + Linear.
    - Duration is predicted in the log domain.
    - Pitch and energy are quantised into 256 bins whose embeddings are added to the hidden sequence.
    - The **length regulator** expands each phoneme's hidden state by its duration.
- **Mel decoder:** 6 FFT blocks (the reference implementation's setting; the paper uses 4), a linear projection to 80 mel bins, and a 5-layer convolutional **PostNet** with a residual connection.
- **Loss:** L1 on the mel spectrogram before and after the PostNet, plus MSE on log-duration, pitch and energy, all with padding masked.
- **Optimiser:** Adam (β=0.9/0.98, $\varepsilon=10^{-9}$) with a Noam warm-up of 4,000 steps, peaking at 9.9×10^{-4}, and step-wise annealing. Gradients are clipped at norm 1.0.
- The **HiFi-GAN** generator and its universal checkpoint are taken from the same repository.

As in the reference implementation, pitch and energy are predicted at **phoneme level**: frame values are averaged over each phoneme, and F0 is linearly interpolated across unvoiced frames. The paper's continuous-wavelet pitch spectrogram is not used, because direct phoneme-level pitch is more stable with only 5 hours of data. The model has 35.1 M parameters.

1.2 Modifications

Area	Change	Reason
Vocabulary	Built from each language frontend's phone inventory and passed to the model; stored in every checkpoint.	Independent English and Telugu models; Telugu support.
English text	Rule-based normalisation of times, dates, years, ordinals, money, percentages, decimals and acronyms. g2p_en applied to the whole sentence (POS-based homograph choice). MFA lexicon generated with the same g2p_en, listing both pronunciations of homographs.	Train/inference consistency; the homograph "conduct" in E2.
Telugu text	New frontend: NFC normalisation, ZWJ/ZWNJ removal, Telugu-digit and number expansion, and a rule-based G2P over akshara structure (Section 2.3).	No Telugu support in the reference implementation.
Data split	Hours-budgeted, speaker-stratified, seeded selection of 5 h train / 1 h validation. No utterance appears in both splits (asserted in code).	Assignment requirement.
Normalisation statistics	Pitch and energy means and standard deviations computed on the **training split only**.	Avoids validation leakage.
Alignment	Utterances whose alignment contains an unknown word (`spn`) are dropped, not turned into pauses.	Prevents mis-aligned duration targets.
Feature extraction	Re-implemented in numpy/scipy: same Slaney mel filterbank, log-clamp at 10^{-5}, energy = L2 norm of each STFT frame. New TextGrid reader.	Fewer dependencies; can be tested without PyTorch.
Length regulator	`torch.repeat_interleave` (identical output), plus a guard against empty output at inference.	Speed and robustness.
Training loop	Resumable training, a wall-clock limit, best-validation checkpoint, full-set validation, guard against NaN/Inf losses, CSV logs, automatic loss plots, and an fp16 inference export.	Colab session limits; reproducibility.
Inference	Test-set mode: WAV files, duration/pitch/energy plots, and `predictions.json` holding the model's raw predictions.	Deliverables are generated directly from model outputs.
Configuration	One `config.yaml` per language, with paths relative to the config file.	Required repository structure.

2. Datasets, preprocessing and training configuration

2.1 Data selection and train/validation split

	English	Telugu
Corpus	Svarah (ai4bharat/Svarah, CC BY 4.0): 6,656 utterances of Indian-accented read English, ~9.6 h, 117 speakers	IndicTTS Telugu (SPRINGLab/IndicTTS_Telugu): 8,576 utterances, 48 kHz, one female and one male speaker
Text filter	Transcripts with annotation brackets removed (0 removed)	Transcripts containing Latin letters removed (code-mixed English words cannot be spelled by a Telugu G2P) (43 removed)
Duration filter	1–11 s per utterance (keeps every utterance under the model's 1,000-frame limit)	
Speaker policy	All speakers. Svarah has no speaker-ID column, so speakers were identified by their demographic profile, giving 115 speakers with 6.76 h after filtering (median 2.6 min, maximum 17.8 min per speaker). A speaker embedding is used.	`auto` policy: use one speaker only if it alone covers the 6.24 h budget (6 h × 1.04). After filtering, the male speaker had 6.21 h and the female 1.89 h. The male fell just short, so both were kept with a speaker embedding (most of the data is male). Inference uses the male speaker.
Sampling	Each speaker contributes in proportion to its available audio. Utterances are shuffled with seed 1234 until 5 h + 1 h (×1.04 margin) is reached. Within each speaker, the first ~1/6 of the selected audio goes to validation, so every validation speaker is also seen in training but no utterance is shared.	
Selected (after trimming)	3,686 train utterances (5.15 h) / 835 validation (1.14 h)	2,743 train utterances (4.50 h) / 563 validation (0.90 h). IndicTTS recordings carry long leading and trailing silences, so trimming removes about 15% of the selected 6.24 h.
Usable after alignment and feature checks	**3,683 train (4.82 h) / 835 validation (1.06 h)**; 3 dropped (2 with no voiced frames, 1 too short)	**2,724 train (4.30 h) / 558 validation (0.86 h)**; 24 dropped because MFA could not align them

Hours after alignment are shorter than after trimming because MFA removes leading and trailing silence inside each utterance.

2.2 Preprocessing pipeline (identical for both languages)

1. **Audio loading and validation:** decoded with soundfile; undecodable clips and clips shorter than 0.3 s after trimming are dropped and counted.
2. **Resampling** to 22,050 Hz mono (polyphase filtering), the vocoder's sampling rate. IndicTTS audio is 48 kHz; Svarah audio is resampled from its native rate.
3. **Silence and loudness:** leading and trailing audio more than 40 dB below the peak frame is trimmed (with a 50 ms margin), and the waveform is peak-normalised to 0.95.
4. **Text normalisation and G2P** with the language frontend. Punctuation becomes the pause token `sp`.
5. **Forced alignment** with the Montreal Forced Aligner (MFA):
    - English: the pretrained `english_us_arpa` acoustic model with the g2p_en lexicon (beam 100, retry beam 400, for accented speech).
    - Telugu: no pretrained Telugu model is assumed, so an MFA acoustic model (monophone → triphone → LDA → SAT) was trained on the selected corpus with the G2P lexicon, then used to align it.

6. **Durations:** phone intervals are converted to mel-frame counts from rounded boundaries, so the sum equals the mel length. Leading and trailing silences are cut; internal silences become `sp`.
7. **Mel spectrogram:** 1,024-point STFT, hop 256 (11.6 ms), Hann window, 80 Slaney mel bins from 0 to 8 kHz, log(max(·, 10^{-5})).
8. **Pitch:** PyWorld DIO + StoneMask at hop resolution; unvoiced frames are linearly interpolated, values are averaged per phoneme and z-normalised. **Energy:** L2 norm of each magnitude-STFT frame, averaged per phoneme and z-normalised.
9. **Vocabulary:** `_` (padding), `sp`, then the full phone inventory. English has 71 symbols (69 stressed ARPAbet phones), all seen in training. Telugu has 52 symbols (50 phones).

English pitch statistics (training split): mean 177.1 Hz, standard deviation 57.4 Hz; energy mean 52.7, standard deviation 38.0. Telugu: pitch mean 194.2 Hz, standard deviation 43.6 Hz; energy mean 50.7, standard deviation 33.0. Four rare Telugu phones (`dz, rr, ts, zh`, i.e. చే, ఱ, జే/ఴ, ఐ) never occur in the training data; none of them appears in the test sentences.

2.3 Telugu-specific text processing

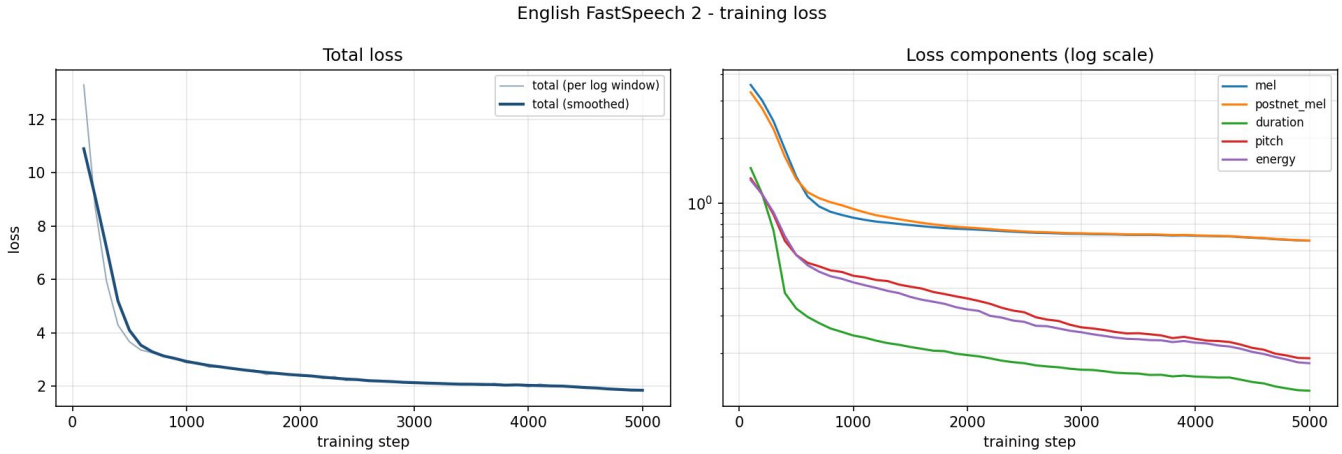
- **Unicode:**
  - NFC normalisation.
  - Zero-width joiner and non-joiner are removed, since they affect glyph shaping, not pronunciation.
  - Telugu digits ౦–౯ are mapped to 0–9, and the danda to a full stop.
- **Numbers to Telugu words:**
  - Cardinals up to crores, using oblique forms (వందల, వేల, లక్షల).
  - The ordinal suffix -వ attaches to the stemmed cardinal (12వ → పన్నెండవ).
  - Times: 10:45 → పది నలభై ఐదు; 2026 → రెండు వేల ఇరవై ఆరు.
- **G2P over akshara structure:**
  - A consonant followed by virama becomes a bare consonant, as in conjuncts and geminates (పట్టు → `p a tt tt u`).
  - A consonant followed by a vowel sign takes that vowel.
  - A bare consonant keeps the inherent *a*, including word-finally, since Telugu has no schwa deletion.
  - The anusvara assimilates to the place of articulation of the next consonant (బలంగా → `b a l a ng g aa`, ఉంచండి → `u ny ch a nn dd i`); otherwise it is *m*.
  - The visarga becomes *h*.
  - Aspiration, retroflexion, vowel length and gemination are kept as distinct phones because they are contrastive in Telugu.
- **Test-sentence fidelity:** the PDF's text layer garbles Telugu (it extracts వచ్చ ింది for the rendered వచ్చింది). The sentences were therefore transcribed from the rendered glyphs and checked to contain exactly the same Telugu characters as the text layer. The space inside అడ విలో (T3) is kept as provided.

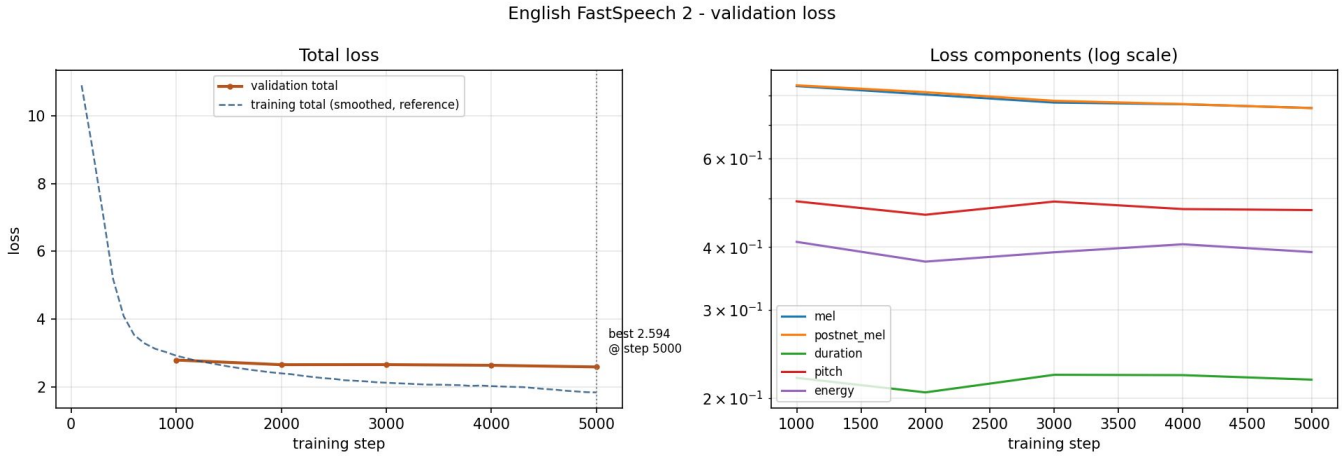
2.4 Training configuration

Parameter	Value (both languages)
Batch size	16, length-bucketed (groups of 4 batches sorted by length)
Learning rate	Noam schedule, 4,000 warm-up steps, peak 9.9×10^{-4}, anneal ×0.3 at 20k/30k/35k
Steps	**5,000 (English) and 3,000 (Telugu)**; 40,000 configured. Training was stopped early because of the submission deadline: Telugu MFA training alone took about 2 hours. The checkpoint with the best validation loss is used for inference (English step 5,000, Telugu step 2,000).
Validation	Full validation set every 1,000 steps; training losses logged every 100 steps
Hardware	Google Colab, NVIDIA T4 (16 GB). English: 38 min for 5,000 steps (~0.46 s/step including validation). Telugu: 26 min for 3,000 steps (~0.52 s/step including validation).
Seed	1234 (data selection, batching, initialisation)

3. Training and validation loss analysis

English: loss curves





English validation losses (full validation set):

Step	Total	Mel L1	PostNet mel L1	Pitch MSE	Energy MSE	Duration MSE
1000	2.796	0.836	0.839	0.493	0.409	0.220
2000	2.660	0.804	0.813	0.464	0.374	0.205
3000	2.662	0.775	0.781	0.492	0.390	0.223
4000	2.641	0.769	0.770	0.476	0.405	0.222
5000	2.594	0.756	0.755	0.474	0.391	0.218

Convergence. The training loss drops quickly from 13.29 at step 100 to about 3 by step 1,000, then decreases steadily to 1.83 at step 5,000. Validation loss falls monotonically except for a negligible step at 3,000 (2.660 → 2.662): 2.796 → 2.659 → 2.662 → 2.641 → 2.594. The best checkpoint is the last one, so the model was still improving when training stopped. The learning rate peaked at step 4,000, and the largest validation drop (−0.048) comes in the last 1,000 steps. Longer training would very likely keep reducing the loss.

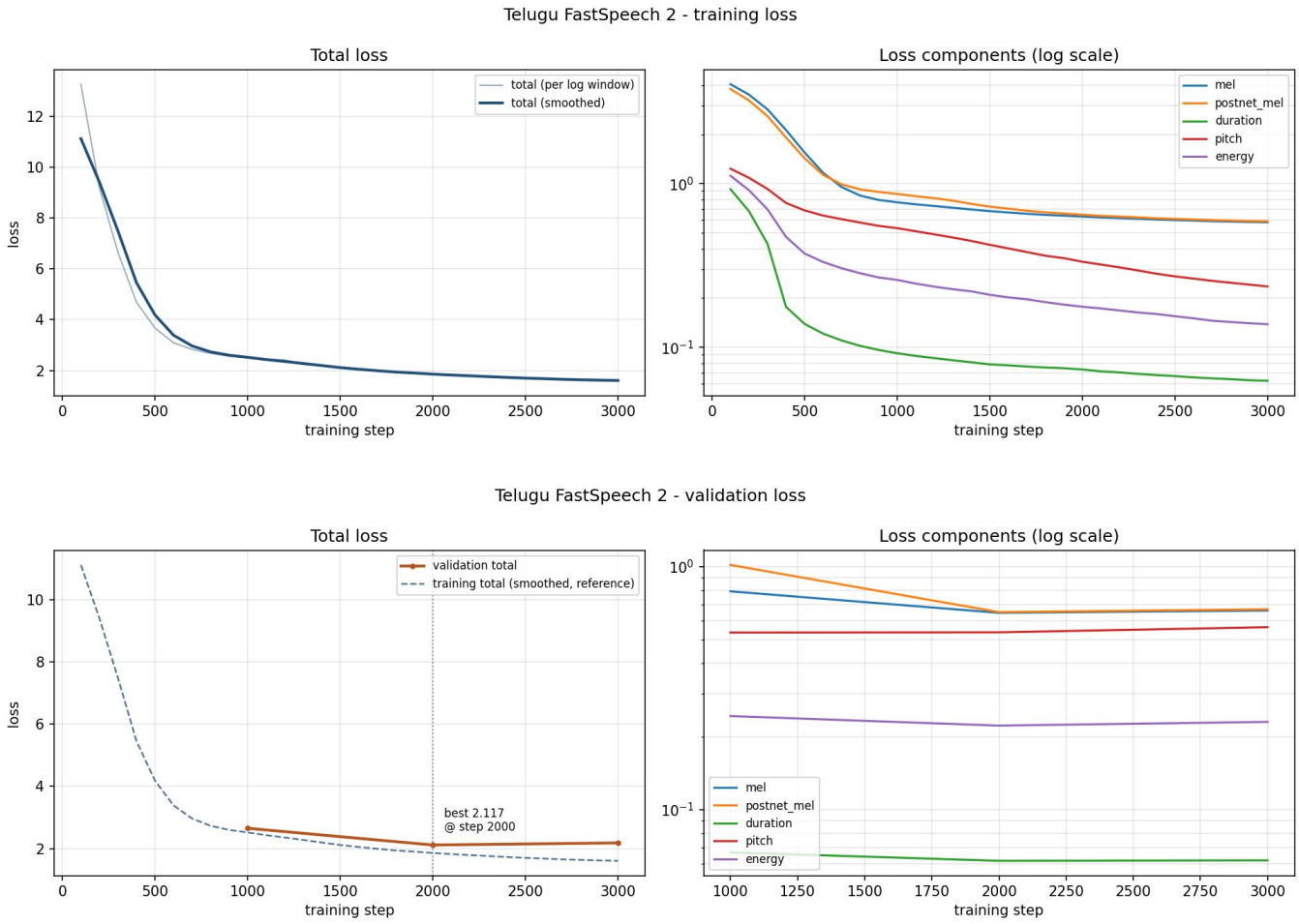
Stability. No NaN or Inf losses or gradients occurred; the guard logged 0 skipped batches. The training curve is smooth, with no spikes, even around the learning-rate peak.

Overfitting. The gap between training and validation loss (1.83 vs. 2.59 at step 5,000) is driven almost entirely by the **variance predictors**:

- Training pitch loss fell to 0.19 while validation pitch loss stayed flat at 0.46–0.49 from step 1,000.
- Energy behaved similarly: 0.18 in training against 0.37–0.41 in validation.
- Duration: 0.13 against 0.21–0.22.
- The mel losses keep improving on validation (0.836 → 0.756), so the decoder is not overfitting.

This early sign of overfitting in the prosody predictors is expected for Svarah. Its 115 speakers average about 2.6 minutes of speech each, so the predictors can memorise speaker- and utterance-specific prosody faster than they learn rules that generalise. Svarah's spontaneous-style, accented read speech is also less predictable than studio TTS data.

Telugu: loss curves



Telugu validation losses (full validation set):

Step	Total	Mel L1	PostNet mel L1	Pitch MSE	Energy MSE	Duration MSE
1000	2.653	0.792	1.017	0.536	0.243	0.067
2000	2.117	0.646	0.650	0.537	0.222	0.062
3000	2.183	0.660	0.668	0.564	0.230	0.062

Convergence. The training loss falls from 13.26 at step 100 to 1.60 at step 3,000, lower than English reached in 5,000 steps. The validation loss drops sharply from 2.653 at step 1,000 to **2.117 at step 2,000**. Most of that drop comes from the PostNet, whose mel L1 falls from 1.017 to 0.650 once it starts to help. At step 3,000 the validation loss rises to 2.183:

- mel 0.646 → 0.660;
- PostNet mel 0.650 → 0.668;
- pitch 0.537 → 0.564;
- while the training loss keeps falling.

Overfitting or noise. This is either the start of overfitting or noise from the still-rising learning rate: at step 3,000 the warm-up has not finished (7.4×10^{-4}). With a single later point the two cannot be separated. The best-validation checkpoint (step 2,000) is therefore used for inference.

Stability. No non-finite losses occurred (0 skipped batches).

Duration. The Telugu duration loss is very low (validation 0.062 against English 0.218), because the data is consistent studio speech dominated by one speaker.

4. Generated speech and variance-adaptor analysis

All audio was generated from each model's best checkpoint (English: step 5,000; Telugu: step 2,000) with HiFi-GAN, using predicted (not ground-truth) durations, pitch and energy. Every plot is drawn from that sentence's own model outputs, which are stored in `predictions.json`:

- **Duration plot:** bars are the rounded frame counts used by the length regulator; dots are the un-rounded predictor output exp(log d)−1.
- **Pitch and energy plots:** phoneme-level predictions, de-normalised to Hz and energy units and drawn over each phoneme's predicted time span.
- The grey curve in the pitch plots is F0 re-extracted from the synthesised waveform. It is a check that the vocoder actually realises the predicted pitch.

Because training F0 targets are interpolated through unvoiced frames, the model also predicts F0 values for voiceless consonants (/s/, /t/, /k/). These values are not audible; they act as a smooth pitch contour that continues between vowels.

English test sentences

Summary of the model's predictions (from `Results/English/predictions.json`):

ID	Phones	Length (s)	Phones/s	Pauses	Pred. F0 range (Hz)	Mean F0 first → last quarter	Pred. energy range
E1	87	5.96	14.6	0	118–173	157 → 132	12–134
E2_a	51	3.45	14.8	0	113–162	156 → 125	10–156
E2_b	56	3.84	14.6	0	120–175	152 → 129	10–128
E3	37	2.65	14.0	0	118–164	149 → 123	12–146
E4	82	6.79	11.9	1	109–201	166 → 136	5–146

Overall listening impression (English). One listener judged all five English outputs:

- **E4** is almost clear and understandable.
- **E1, E2_a, E2_b and E3** sound too fast and are hard to understand; only a few words can be made out.

The predictions explain this difference:

- E1–E3 run at 14.0–14.8 phones/s with no pauses.
- E4 runs at 11.9 phones/s and contains the only predicted pause (557 ms).

Two causes follow from this:

1. The **duration predictor** compresses phones too much. Many consonants and unstressed vowels get only 3–5 frames (35–58 ms), so syllables run together.
2. At 5,000 steps the **decoder** has not yet learnt sharp phone transitions: the validation mel L1 is still falling (0.756). Its output is a smoothed, averaged spectrogram, which HiFi-GAN renders as slurred speech.

Both are expected this early in training: the paper trains for 160k steps on a single speaker. Here the model has 115 speakers with about 2.6 minutes each. Intelligibility should improve with longer training. At inference, FastSpeech 2's duration control (`--duration_control 1.2`) can slow the speech without retraining.

Patterns common to all English sentences.

1. **Declination:** in every sentence the predicted F0 falls from the first to the last quarter (157→133, 156→125, 152→129, 149→122 and 166→135 Hz), the normal falling intonation of declaratives.
2. **Phrase-final lengthening:** the last one or two phones are among the longest in each sentence. Examples: final /z/ of "proceedings" 13 frames (151 ms), /s/ of "audience" 14 frames, /m/ of "time" 15 frames.
3. **Stress-related lengthening:** stressed vowels are longer than unstressed ones in four of five sentences (E1 65 vs. 54 ms, E2_a 71 vs. 55, E3 81 vs. 61, E4 76 vs. 60; E2_b about equal).
4. **Energy:** vowels carry about twice the energy of consonants (around 100 vs. 50–60). Energy dips sharply at stop closures (e.g. /k/ and /t/ around 10–18) and decays at the end of each sentence.
5. **Speaking rate** is consistent at about 14–15 phones/s (13.0 in E4 excluding its pause).
6. The F0 re-extracted from the audio follows the predicted phoneme-level F0 closely in all sentences. Deviations appear only at voicing onsets and offsets.

E1

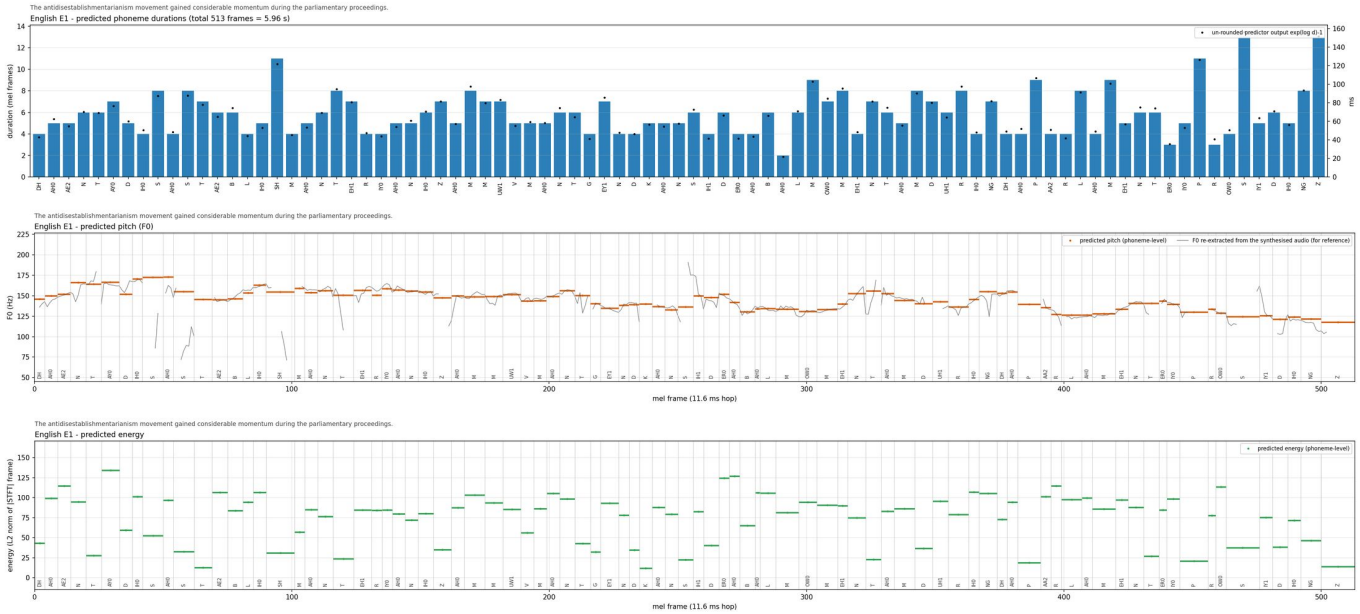
The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

Phone input: DH AH0 AE2 N T AY0 D IH0 S AH0 S T AE2 B L IH0 SH M AH0 N T EH1 R IY0 AH0 N IH0 Z AH0 M M UW1 V M AH0 N T G EY1 N D K AH0 N S IH1 D ER0 AH0 B AH0 L M OW0 M EH1 N T AH0 M D UH1 R IH0 NG DH AH0 P AA2 R L AH0 M EH1 N T ER0 IY0 P R OW0 S IY1 D IH0 NG Z

Listening observations. Perceived as too fast and hard to understand; only a few words can be made out. The 28-phone word is not recognisable as a whole.

Variance-adaptor analysis. 87 phones, predicted 5.96 s at 14.6 phones/s. The sentence has no internal punctuation and the model predicts no pause, so it is synthesised as a single intonation phrase.

- **Pitch:** F0 starts highest early in the sentence (peak 173 Hz on a vowel inside "antidisestablishmentarianism") and declines to about 120 Hz by "parliamentary proceedings" (first-quarter mean 157 Hz, last 133 Hz). The long word's 28 phones come from g2p_en's neural model (`AE2 N T AY0 D IH0 S AH0 S T AE2 B L IH0 SH M AH0 N T EH1 R IY0 AH0 N IH0 Z AH0 M`). Its primary stress EH1 is not marked by higher F0: stressed vowels average 142 Hz against 148 Hz for unstressed ones, so at 5,000 steps lexical stress is expressed mainly by duration and energy, not pitch.
- **Duration:** the longest phones are fricatives: final /z/ 13 frames, /s/ 13, /ʃ/ 11.
- **Energy:** vowels average 96.5 and consonants 60.5. Energy falls to 13.6 on the final /z/.



E2_a

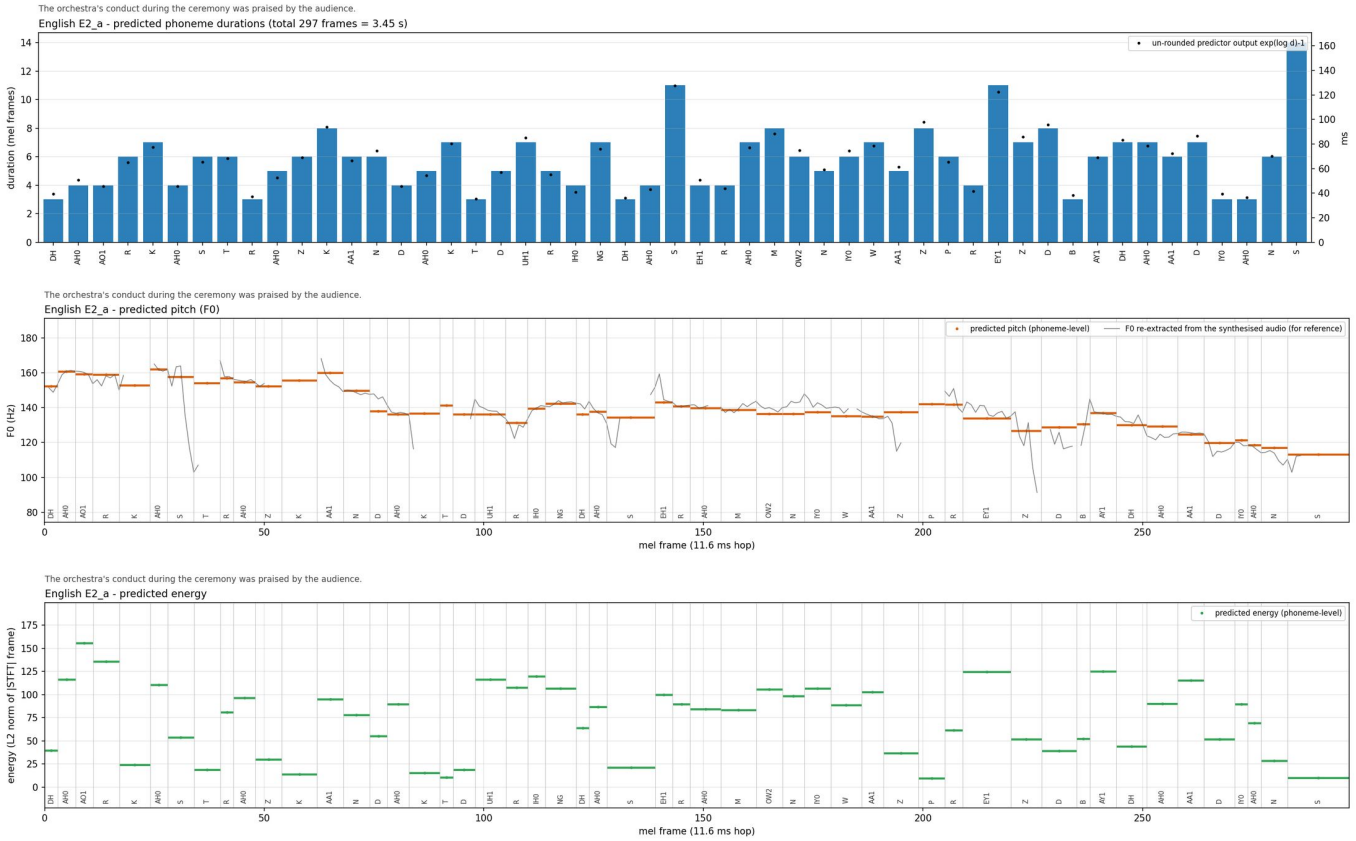
The orchestra's conduct during the ceremony was praised by the audience.

Phone input: DH AH0 AO1 R K AH0 S T R AH0 Z K AA1 N D AH0 K T D UH1 R IH0 NG DH AH0 S EH1 R AH0 M OW2 N IY0 W AA1 Z P R EY1 Z D B AY1 DH AH0 AA1 D IY0 AH0 N S

Listening observations. Perceived as too fast and hard to understand; only a few words can be made out, so the noun/verb stress contrast with E2_b cannot be confirmed by ear, although it is visible in the predictions (below).

Variance-adaptor analysis. Homograph test (noun). g2p_en selects the noun pronunciation of "conduct", `K AA1 N D AH0 K T`, with stress on the first syllable.

- **Pitch:** the model puts the local F0 peak on the stressed AA1 (160 Hz) and falls through the second syllable (D AH0 ≈136 Hz), a falling, first-syllable-prominent contour.
- **Energy:** AA1 95 vs. AH0 90.
- **Duration:** "praised" carries the longest vowel (EY1, 11 frames), and the final /s/ of "audience" is lengthened to 14 frames (163 ms), the longest phone in the sentence.
- **Overall F0:** declines from 156 to 125 Hz across the sentence.



E2_b

The researchers will conduct an experiment to evaluate the proposed method.

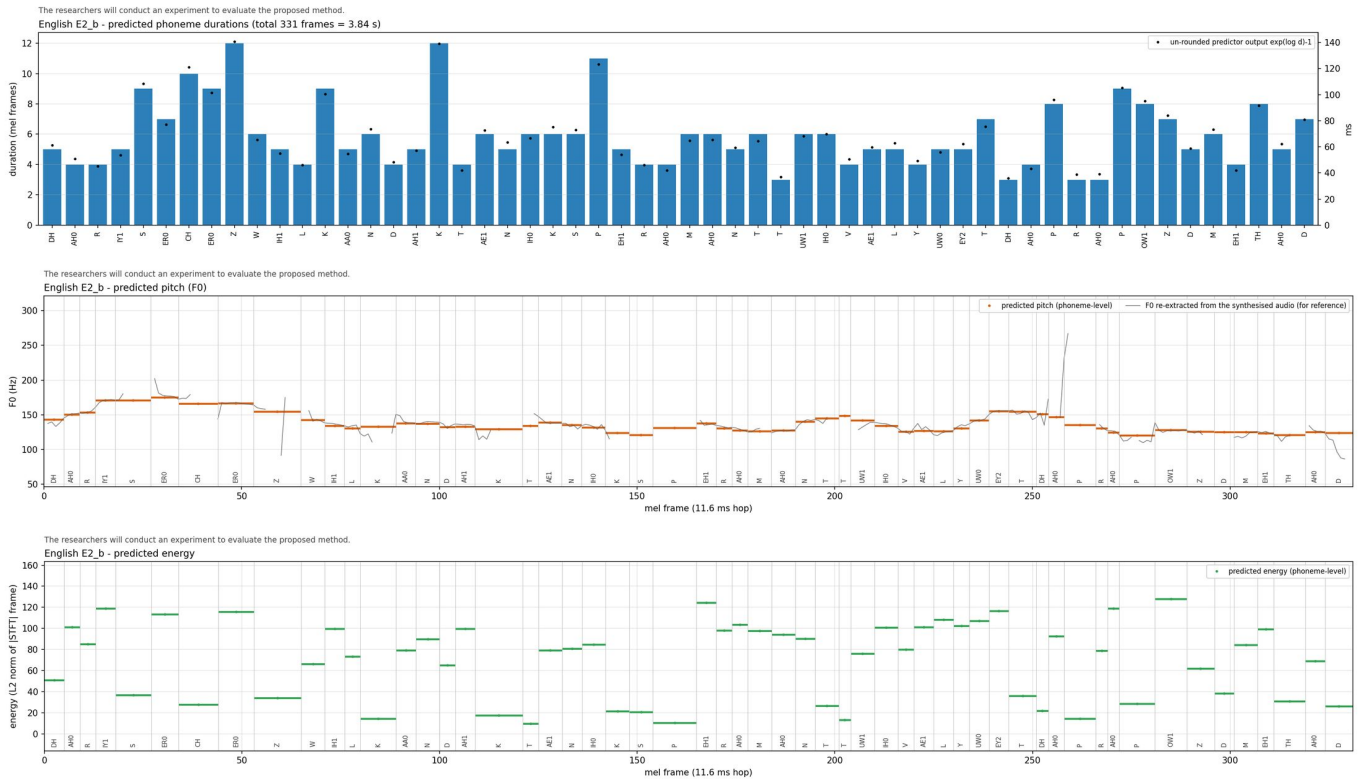
Phone input: `DH AH0 R IY1 S ER0 CH ER0 Z W IH1 L K AA0 N D AH1 K T AE1 N IH0 K S P EH1 R AH0 M AH0 N T T UW1 IH0 V AE1 L Y UW0 EY2 T DH AH0 P R AH0 P OW1 Z D M EH1 TH AH0 D`

Listening observations. Perceived as too fast and hard to understand; only a few words can be made out.

Variance-adaptor analysis. Homograph test (verb). The same spelling gets the verb pronunciation `K AA0 N D AH1 K T`, with stress on the second syllable, so the frontend disambiguates E2_a and E2_b correctly from context.

- The variance predictors respond to the stress change. The first-syllable F0 peak of the noun disappears: the verb's AA0 is at 138 Hz versus 160 Hz in E2_a, giving a flat 129–138 Hz contour over the word.
- Prominence moves to the second syllable mainly through **energy**: AH1 is at 100 against 79 for AA0, while in the noun the first vowel was the stronger one.
- The closure of the word-final /k/ of "conduct" is long (12 frames) before "an experiment".

Pitch moves less than the dictionary stress would suggest: in this read, accented English the model realises the noun/verb distinction more through energy and vowel quality than through pitch.



E3

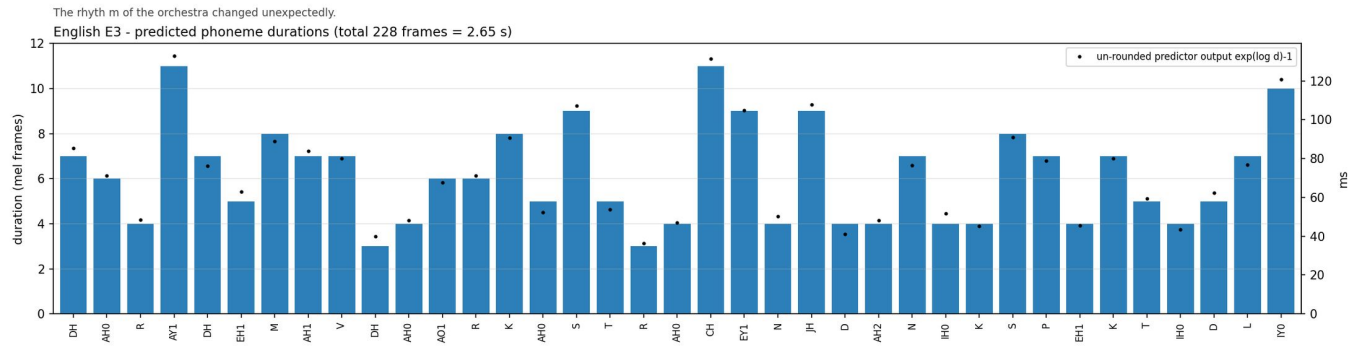
The rhyth m of the orchestra changed unexpectedly.

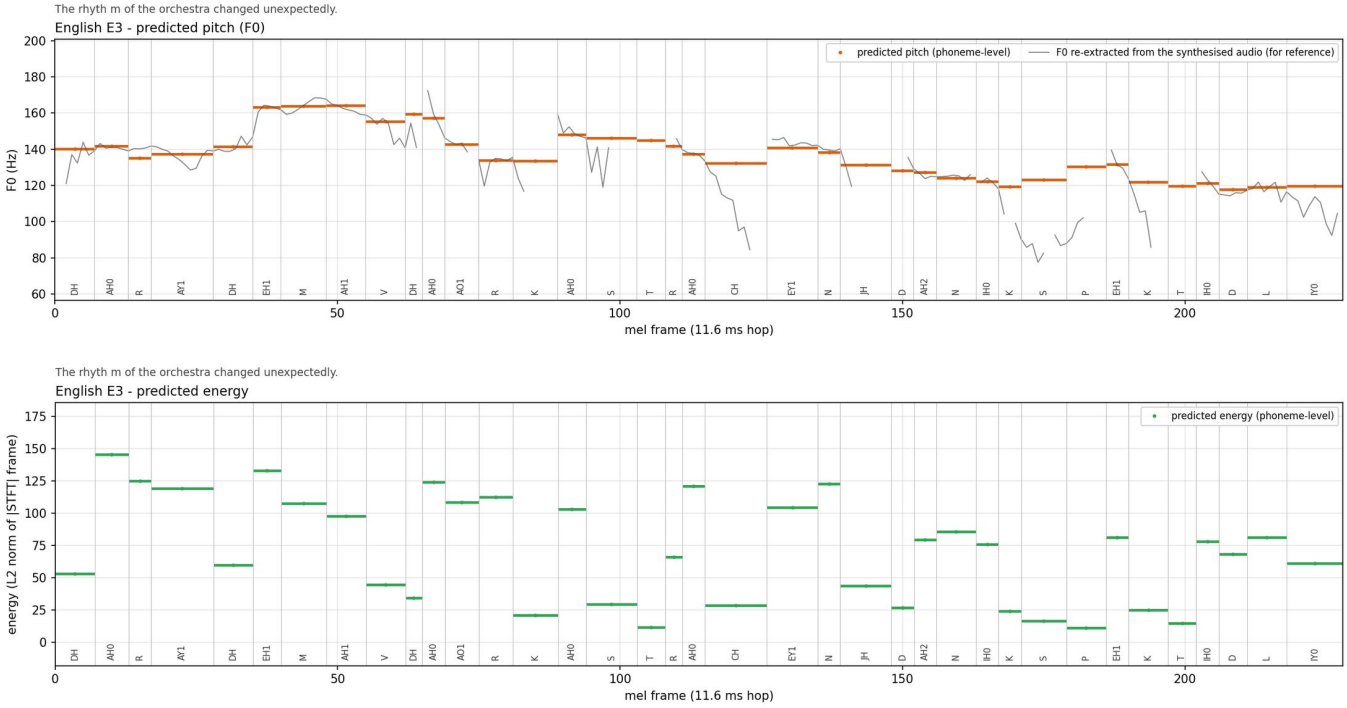
Phone input: `DH AH0 R AY1 DH EH1 M AH1 V DH AH0 AO1 R K AH0 S T R AH0 CH EY1 N JH D AH2 N IH0 K S P EH1 K T IH0 D L IY0`

Listening observations. Perceived as too fast and hard to understand; only a few words can be made out.

Variance-adaptor analysis. The test text contains "rhyth m" with an internal space, kept as written. The frontend therefore sees two words: g2p_en pronounces the non-word "rhyth" as `R AY1 DH` ("rythe") and the single letter "m" as its letter name `EH1 M` ("em"). The model reproduces this faithfully, so the output says "rythe em", not "rhythm".

- **Pause:** no pause is predicted at the space, which is correct, because a space is not punctuation.
- **Duration:** AY1 is one of the longest vowels (11 frames), the /tʃ/ of "changed" is 11 frames, and the final IY0 of "unexpectedly" is lengthened to 10 frames. This sentence has the clearest stress-lengthening of the set: stressed vowels average 81 ms against 61 ms for unstressed ones.
- **Pitch:** F0 declines from 149 to 122 Hz, with the peak on "of" (164 Hz).
- **Energy:** energy stays high to the end (61 on the final vowel), because the sentence ends in a vowel and not a voiceless consonant.





E4

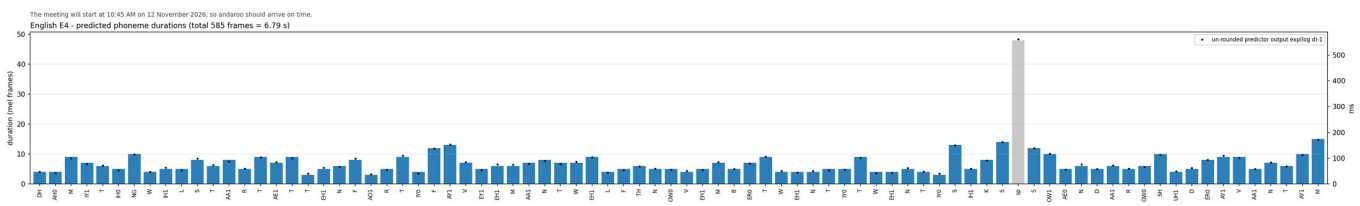
The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Phone input: `DH AH0 M IY1 T IH0 NG W IH1 L S T AA1 R T AE1 T T EH1 N F AO1 R T IY0 F AY1 V EY1 EH1 M AA1 N T W EH1 L F TH N OW0 V EH1 M B ER0 T W EH1 N T IY0 T W EH1 N T IY0 S IH1 K S sp S OW1 AE0 N D AA1 R OW0 SH UH1 D ER0 AY1 V AA1 N T AY1 M`

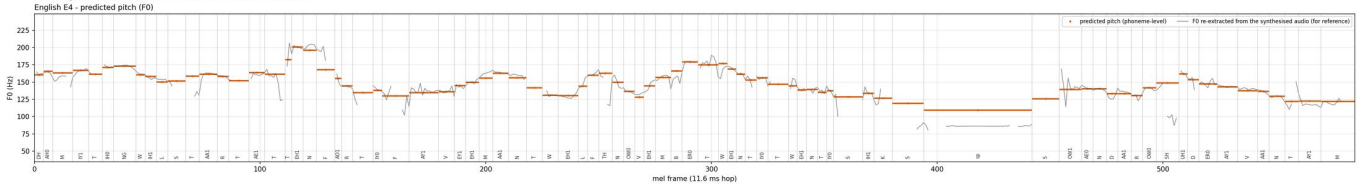
Listening observations. Almost clear and understandable — the most intelligible English output. It is also the slowest sentence (11.9 phones/s) and the only one with a predicted pause, which supports the view that speaking rate is the main factor limiting intelligibility at this stage of training.

Variance-adaptor analysis. This sentence tests text normalisation.

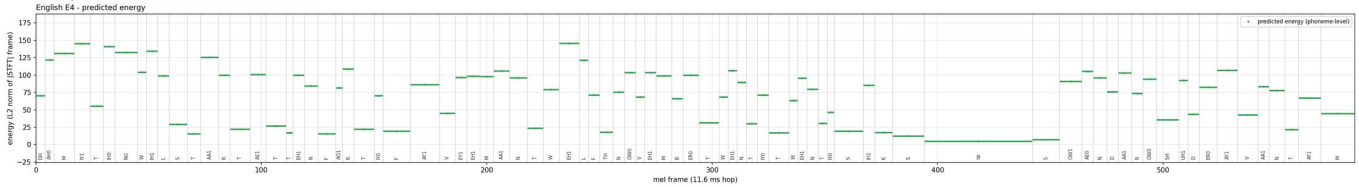
- **Normalisation:** "10:45 AM" becomes "ten forty five A M" (`EY1 EH1 M`) and "12 November 2026" becomes "twelfth November twenty twenty six". "andaroo", a Telugu word written in Latin script, is pronounced by g2p_en as `AE0 N D AA1 R OW0` .
- **Pause:** the comma after "2026" produces the only pause in the English test set, predicted at **48 frames (557 ms)**, the longest duration in any English plot. The model has learned punctuation pauses from the MFA silences. Because of this pause, the overall rate is 11.9 phones/s (13.0 excluding it).
- **Pitch:** the highest F0 of the whole English test set (201 Hz) falls on "ten", the start of the time expression, a prominence typical of new numeric information. F0 resets slightly after the pause and then declines again to the end (166 Hz first quarter, 135 Hz last quarter).
- **Duration:** the final /m/ of "time" is lengthened to 15 frames (174 ms).



The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.



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Telugu test sentences

Summary of the model's predictions (from `Results/Telugu/predictions.json`):

ID	Phones	Length (s)	Phones/s	Pauses	Pred. F0 range (Hz)	Mean F0 first → last quarter	Pred. energy range
T1	66	4.55	14.5	0	112–265	206 → 155	1–147
T2_a	31	2.15	14.4	0	118–250	180 → 170	8–129
T2_b	37	2.35	15.8	0	108–243	183 → 148	6–132
T3	31	1.82	17.0	0	110–239	183 → 132	8–142
T4	120	7.77	15.3	1	108–235	199 → 170	1–132

Overall listening impression (Telugu). The listener rated **sound quality as very good in all sentences**.

- **T4** was judged the best.
- **T1, T2_a, T2_b and T3** are slightly fast but understandable, decently good overall.

The predicted rates agree: T3 is the fastest at 17.0 phones/s, and T4 is the only sentence with a predicted pause. The Telugu model is clearly more intelligible than the English one, even though it trained for fewer steps (Section 5).

Patterns common to all Telugu sentences.

1. **Vowel length:** long vowels (aa, ii, uu, ee, oo) are predicted roughly **twice as long** as short ones: 81–104 ms against 47–57 ms in every sentence. The model learned this phonemic contrast from the G2P's separate long and short vowel symbols.
2. **Gemination:** geminate consonants are realised as long closures. పట్టు `tt tt` lasts 10–11 frames (116–128 ms) and నిశ్శబ్దంగా `sh sh` 18 frames, against 3–7 frames for a single stop. Energy drops to about 10–15 during the closure.
3. **Declination:** F0 falls over every sentence (first → last quarter: 206→155, 180→170, 183→148, 183→132, 199→170 Hz). Energy decays to almost zero at the final vowel, which in all five sentences is the inherent or final *u/i* of a Telugu verb.
4. **Energy:** vowels carry about twice the energy of consonants (72–88 against 35–46).
5. **Anusvara assimilation:** it appears as distinct nasal phones (`ng` before /g/, `ny` before /ch/, `nn` before /dd/, `m` before /b/). Each gets its own short duration of about 4–6 frames.

T1
న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

Phone input: `n y uu r oo ph aa r m a k oo l aa j i k a l p a r i sh oo dh a n a l oo n uu t a n a au ss a dh aa l a p r a bh aa v aa n n i v i sh l ee ss i ny ch aa r u`

Listening observations. Very good sound quality; slightly fast but understandable.

Variance-adaptor analysis. 66 phones, 4.55 s at 14.5 phones/s, no pause.

- **Phonemes:** the 9-akshara loan word న్యూరోఫార్మకోలాజికల్ is converted conjunct by conjunct: `n y uu r oo ph aa r m a k oo l aa j i k a l`. న్యూ becomes /n y uu/, ర్మ becomes /r m/, and the word-final virama gives a bare /l/.

- **Pitch:** the sentence opens with the highest predicted F0 of the Telugu set, 265 Hz on the stressed /oo/ of "nyuuroo". F0 then declines steadily: first-quarter mean 206 Hz, last-quarter 155 Hz.
- **Duration:** long vowels average 91 ms against 47 ms for short ones. The longest phones are the aspirated /ph/ (11 frames) and the /sh/ of విశ్లేషించారు (11 frames). The diphthong of ఔషధాల is a single /au/ token.
- **Energy:** falls to almost zero on the final /u/.



T2_a

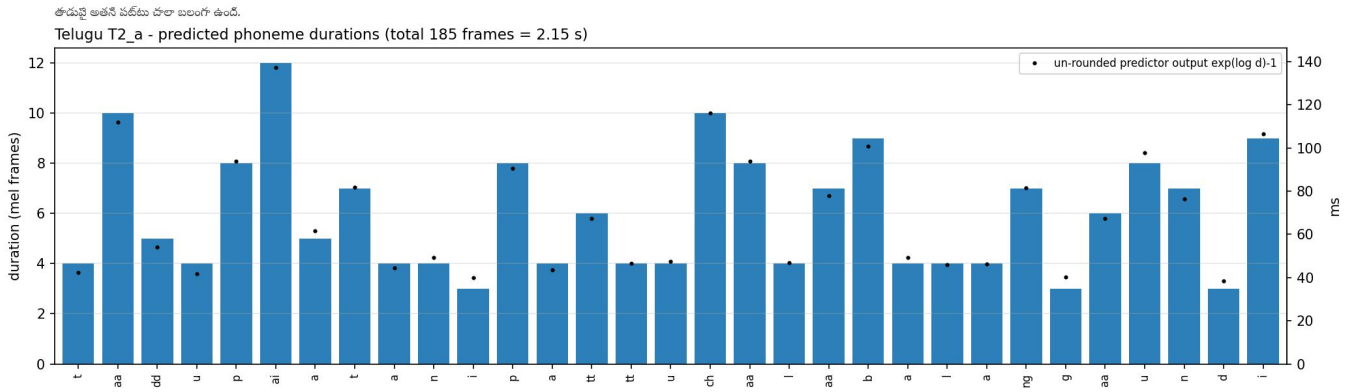
తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

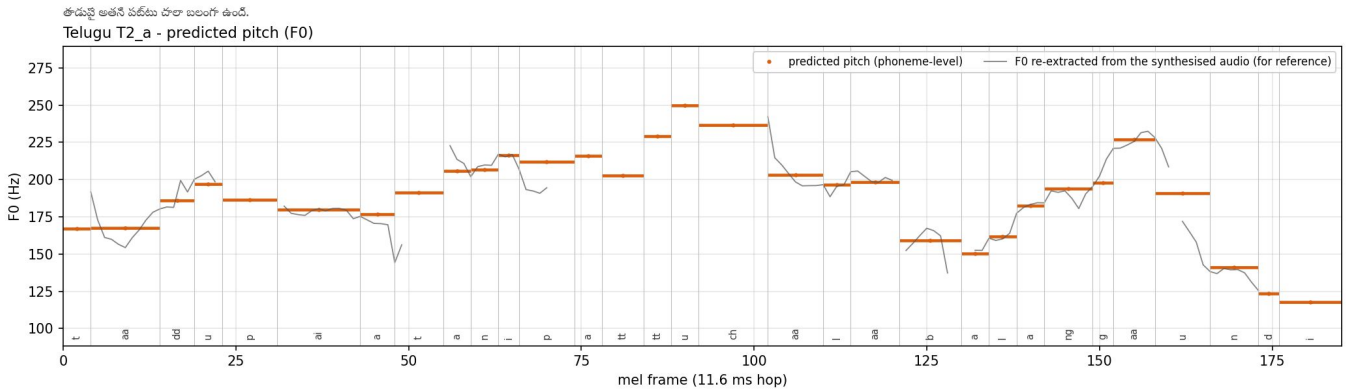
Phone input: `t aa dd u p ai a t a n i p a tt tt u ch aa l aa b a l a ng g aa u n d i`

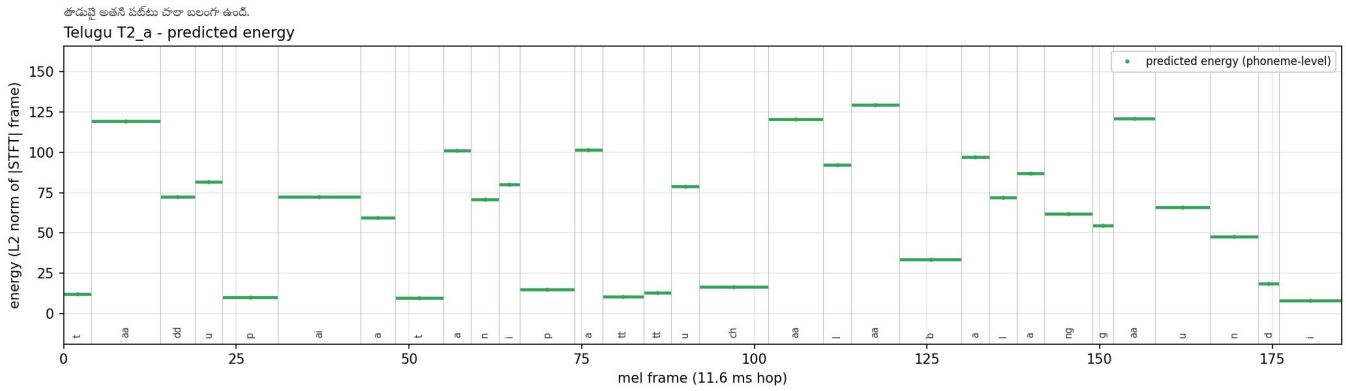
Listening observations. Very good sound quality; slightly fast but understandable.

Variance-adaptor analysis. పట్టు here means "grip". Telugu spells and pronounces the "grip" and "silk" senses identically, so the G2P correctly gives the same phones `p a tt tt u` in T2_a and T2_b; only the context differs.

- **Duration:** the geminate /tt tt/ lasts 10 frames (116 ms), about twice a single stop. The diphthong /ai/ of తాడుపై is the longest phone (12 frames).
- **Anusvara:** in బలంగా it is realised as velar /ng/ before /g/.
- **Pitch:** 31 phones in 2.15 s (14.4 phones/s). F0 stays high and fairly level (180→170 Hz) and peaks on the /u/ of పట్టు (250 Hz), giving the object of the sentence prominence.







T2_b

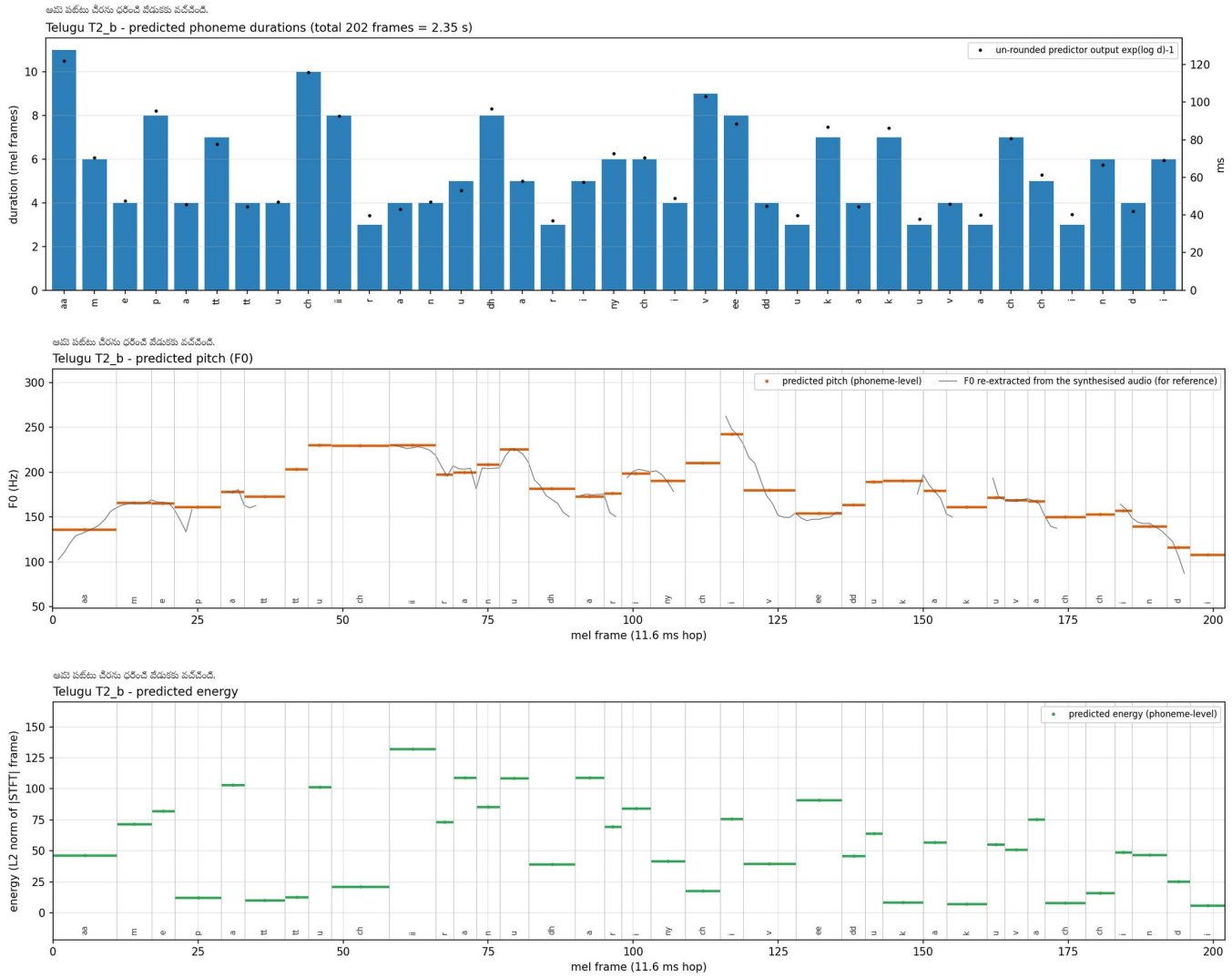
ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

Phone input: `aa m e p a tt tt u ch ii r a n u dh a r i ny ch i v ee dd u k a k u v a ch ch i n d i`

Listening observations. Very good sound quality; slightly fast but understandable.

Variance-adaptor analysis. Same word పట్టు ("silk") with the same phones.

- **Duration:** the geminate /tt tt/ gets 11 frames (128 ms), almost exactly as in T2_a. The model treats the word consistently, as expected for identical pronunciations. The long /ii/ of చీరను and the geminate /ch ch/ of వచ్చింది are lengthened.
- **Anusvara:** in ధరించి it becomes palatal /ny/ before /ch/.
- **Rate:** fast at 15.8 phones/s; short vowels average 47 ms against 104 ms for long ones.
- **Pitch:** declines from 183 to 148 Hz; energy decays to 6 at the final /i/.



T3

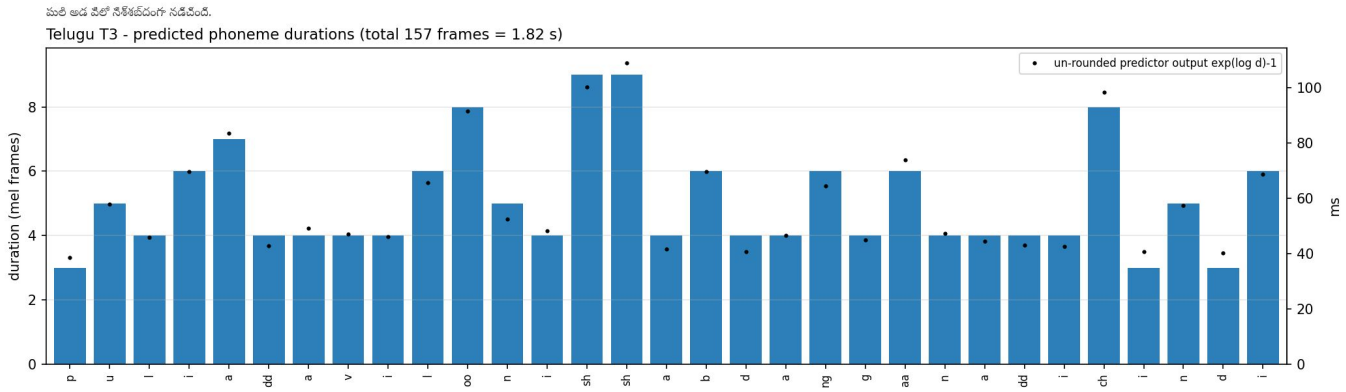
పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

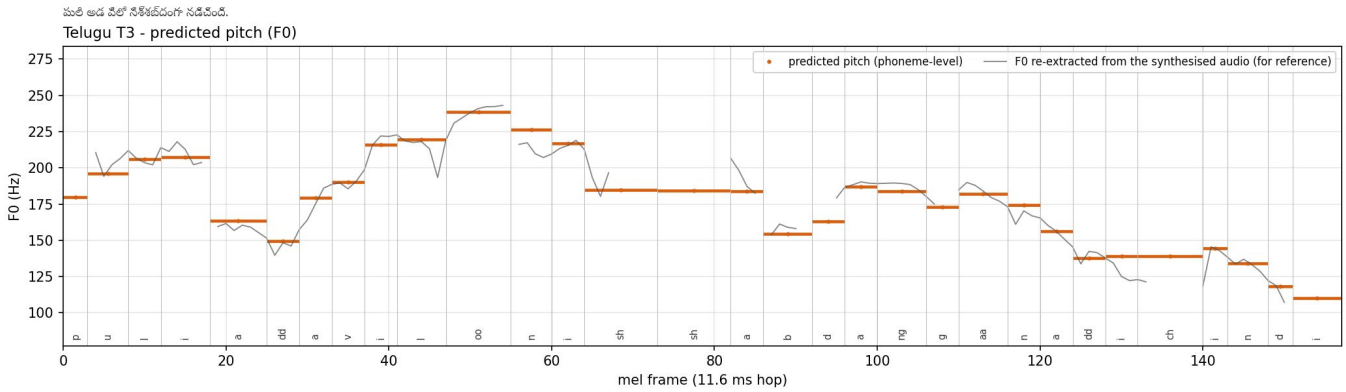
Phone input: `p u l i a dd a v i l oo n i sh sh a b d a ng g aa n a dd i ch i n d i`

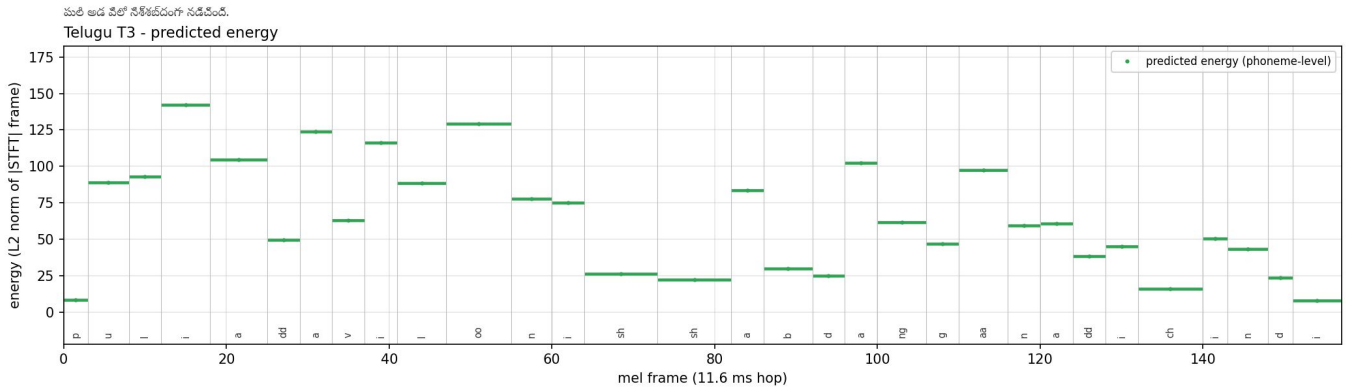
Listening observations. Very good sound quality; slightly fast but understandable.

Variance-adaptor analysis. The test text contains అడ విలో with an internal space, kept as written. The G2P output for the two pieces (`a dd a + v i l oo`) is identical to that of అడవిలో, and the model predicts no pause at the space, so the intended word "in the forest" is produced. Unlike English E3, the space here causes no mispronunciation.

- **Rate:** the fastest Telugu sentence: 31 phones in 1.82 s (17.0 phones/s).
- **Duration:** the longest segment is the geminate /sh sh/ of నిశ్శబ్దంగా (9+9 frames, 209 ms).
- **Anusvara:** in the same word it is velar /ng/ before /g/.
- **Pitch:** the strongest declination of the set, 183→132 Hz, peaking on the /oo/ of విలో.







T4

ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

Phone input: `ii s a m aa v ee sh a m r e nn dd u v ee l a i r a v ai aa r u n a v a m b a r p a n n e nn dd a v a t ee d ii u d a y a m p a d i n a l a bh ai ai d u g a nn tt a l a k u p r aa r a m bh a m a v u t u n d i sp s a m a y aa n i k i s i d dh a ng g aa u ny ch a nn dd i`

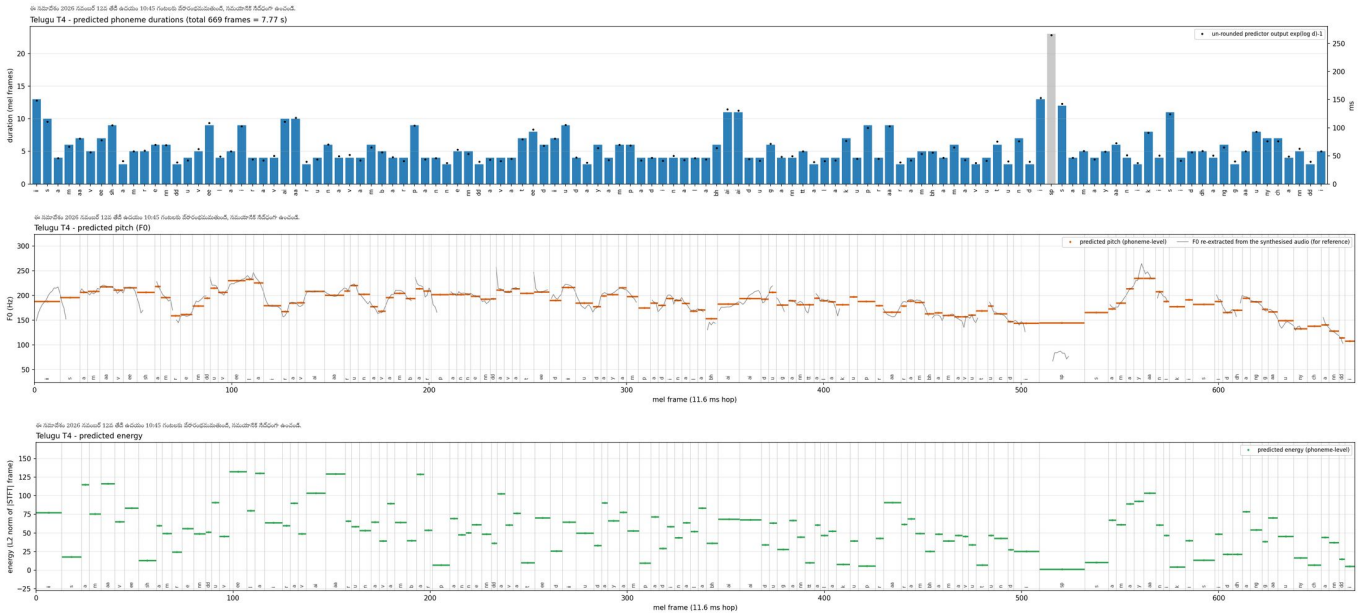
Listening observations. The best Telugu output: very good sound quality and clearly understandable.

Variance-adaptor analysis. This sentence tests number normalisation.

- **Normalisation:** "2026" becomes రెండు వేల ఇరవై ఆరు, "12వ" becomes పన్నెండవ (cardinal stem + ordinal suffix), and "10:45" becomes పది నలభై ఐదు. This gives 120 phones and 7.77 s, the longest output. నవంబర్ correctly ends in a bare /r/, and the anusvara of ప్రారంభమవుతుంది becomes /m/ before /bh/.
- **Pause:** the comma produces the only Telugu pause, 23 frames (267 ms).

- **Duration:** the vowel before the pause (the final /i/ of ప్రారంభమవుతుంది, 13 frames) and the sentence-initial ఈ (13 frames) are the longest phones, showing pre-pausal and initial lengthening.
- **Pitch:** after the pause, F0 resets to its maximum (235 Hz, at సమయానికి) and then falls again: a clear two-phrase intonation (overall 199→170 Hz).

The pause and phrase structure probably explain why T4 sounds best, matching English E4.



5. Comparison of the English and Telugu models; limitations and improvements

5.1 Comparison

	English (Svarah)	Telugu (IndicTTS)
Steps / best checkpoint	5,000 / 5,000	3,000 / 2,000
Best validation loss	2.594	2.117
Validation mel / PostNet L1	0.756 / 0.755	0.646 / 0.650
Validation duration / pitch / energy MSE	0.218 / 0.474 / 0.391	0.062 / 0.537 / 0.222
Speaking rate of test outputs	11.9–14.8 phones/s	14.4–17.0 phones/s
Listening result	Only E4 almost clear; others too fast and hard to follow	Very good quality; all understandable, slightly fast

The Telugu model is much better, despite fewer steps and slightly less data. Four factors explain this:

1. **Data consistency.** IndicTTS is studio TTS speech, 77% of it from one male speaker. Svarah has 115 speakers with about 2.6 minutes each, recorded in varied conditions, with Indian-accented read speech. The duration MSE (0.06 vs. 0.22) and mel L1 (0.65 vs. 0.76) show how much more predictable the Telugu targets are.
2. **Pronunciation lexicon fit.** Telugu orthography is almost phonemic, so the rule-based G2P matches what the speakers say. English uses a US-English ARPAbet lexicon and acoustic model to align Indian-accented speech. Vowel qualities, retroflex stops and stress patterns differ from the dictionary, which makes alignments and duration targets noisier.

3. **Out-of-vocabulary text.** The English test set is designed to break G2P. "antidisestablishmentarianism" (E1), "rhyth m" with a space (E3) and the Telugu word "andaroo" in Latin script (E4) are pronounced by g2p_en's neural model. "rhyth m" becomes "rythe em". In Telugu, the space in అడ విలో (T3) has no phonetic effect: the G2P output equals that of అడవిలో and no pause is predicted, so the model reads the intended word.
4. **Shared behaviour.** Both models learned punctuation pauses (557 ms in E4, 267 ms in T4) and pitch reset after the pause, F0 declination, phrase-final lengthening and the vowel–consonant energy contrast. Both speak too fast, and in both languages the sentence with a comma pause (E4, T4) was judged the best.

5.2 Limitations and failures

- **Very short training** (5,000 and 3,000 steps, against 160k in the paper), forced by the deadline. The English validation loss was still falling.
- **English intelligibility:** E1–E3 are hard to understand. E3 is mispronounced as written ("rythe em") because the test text was preserved exactly. The long word in E1 is not recognisable.
- **Speaking rate:** both duration predictors under-predict the duration of consonants and short vowels, which makes the speech too fast.
- **Prosody generalisation:** the English pitch and energy predictors overfit early (validation flat while training falls). For Telugu the pitch validation loss is the highest component (0.54–0.56).
- **Telugu coverage:**
  - 4 rare phones never occur in training.
  - Code-mixed transcripts (Latin script) were excluded, so the model cannot read English words in Telugu text.
  - 24 utterances could not be aligned.
- **Vocoder:** the universal HiFi-GAN was trained on English. It performed well for Telugu, but it is not fine-tuned to either voice.
- **Speaker identities in Svarah** are derived from demographic metadata and may merge two speakers.

5.3 Possible improvements

1. **Train longer** (the configured 40k steps or more), selecting the checkpoint by validation loss. This is the single largest expected gain, especially for English.
2. **English data:** train on the few Svarah speakers with the most audio, or use speaker-balanced batches. Train an MFA acoustic model on Svarah itself, as was done for Telugu, and add an Indian-English pronunciation lexicon to reduce alignment errors.
3. **Speaking rate:** use inference-time duration control (`--duration_control 1.1–1.2`), or weight the duration loss more heavily.
4. **Telugu:** train a single-speaker male model, which needs only a slightly lower hours budget. Add transliteration for code-mixed English words.
5. **Prosody:** use the paper's CWT pitch spectrogram, frame-level energy, and stronger dropout or early stopping on the pitch and energy predictors.
6. **Vocoder:** fine-tune HiFi-GAN on ground-truth-aligned mels of each language.

6. Reproducibility

Repository: https://github.com/diyamofficial/fastspeech2-english-telugu-tts. Code, configuration files, both models' results and the fp16 inference checkpoints are included. The full training checkpoints are linked from the README.

Software: Python ≥ 3.9, PyTorch ≥ 2.0 (CUDA), numpy, scipy, matplotlib, PyYAML, soundfile, pyworld, g2p_en 2.1.0, nltk, datasets and huggingface_hub (`requirements.txt`). The Montreal Forced Aligner is installed from conda-forge. The universal HiFi-GAN weights come from ming024/FastSpeech2. Svarah requires accepting its access conditions on Hugging Face and logging in with a read token.

Configuration: `English/config.yaml` and `Telugu/config.yaml` hold every dataset, path, audio, model, training and inference parameter. All randomness is seeded (1234).

Commands (from the repository root; `colab/FastSpeech2_Colab.ipynb` runs the same steps on a Colab T4):

```
pip install -r requirements.txt
python tools/smoke_test.py --language both          # 2-minute end-to-end check on synthetic data

python English/preprocess.py --stage all            # prepare -> MFA align -> features
python English/train.py --max_steps 5000            # as used for this report (40k configured)
python English/inference.py --test_set              # WAVs + duration/pitch/energy plots ->
Results/English

python Telugu/preprocess.py --stage all
python Telugu/train.py --max_steps 5000
python Telugu/inference.py --test_set

python tools/verify_deliverables.py                 # checks every required file
```

Outputs:

- `Results/<Lang>/training_loss.png` and `validation_loss.png`, with the logs in `Results/<Lang>/logs/` .
- `test_generated_audios/<ID>.wav` .
- `variance_adaptor_plots/{duration,pitch,energy}/<ID>_<kind>.png` .
- `predictions.json` with the raw predicted values.
- Data-selection summaries.